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Print component with memory array using intermittent clock signal

Abstract

A print component includes a plurality of data pads, a clock pad to receive an intermittent clock signal, and a plurality of actuator groups each corresponding to a different liquid type and to a different one of the data pads. Each actuator group includes a plurality of configuration functions, an array of fluid actuators, and an array of memory elements including a first portion corresponding to the plurality of configuration functions and a second portion corresponding to the array of fluid actuators. Each time the intermittent clock signal is present on the clock pad, the array of memory elements to serially load a segment of data bits from the corresponding data pad, including loading a first portion of data bits into the first portion of memory elements, and loading a second portion of data bits into the second portion of memory elements.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a U.S. Continuation Application of National Stage application Ser. No. 16/767,914, filed May 28, 2020, entitled “PRINT COMPONENT WITH MEMORY ARRAY USING INTERMITTENT CLOCK SIGNAL” which is a U.S. National Stage Application of PCT Application No. PCT/US2019/016727, filed Feb. 6, 2019, entitled “PRINT COMPONENT WITH MEMORY ARRAY USING INTERMITTENT CLOCK SIGNAL”.

BACKGROUND

(1) Some print components may include an array of nozzles and/or pumps each including a fluid chamber and a fluid actuator, where the fluid actuator may be actuated to cause displacement of fluid within the chamber. Some example fluidic dies may be printheads, where the fluid may correspond to ink or print agents. Print components include printheads for 2D and 3D printing systems and/or other high precision fluid dispense systems.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a block and schematic diagram illustrating a print component, according to one example.

(2) FIG. 2 is a block and schematic diagram illustrating a print component, according to one

example.

(3) FIG. 3 is a block and schematic diagram generally illustrating portions of a primitive arrangement, according to one example.

(4) FIG. 4A is a schematic diagram generally illustrating data segments, according to one example.

(5) FIG. 4B is a schematic diagram generally illustrating data segments, according to one example.

(6) FIG. 5 is a block and schematic diagram illustrating a print component, according to one example.

(7) FIG. 6 is a block and schematic diagram illustrating a print component, according to one example.

(8) FIG. 7 is a schematic diagram illustrating a block diagram illustrating one example of a fluid ejection system.

(9) FIG. 8 is a flow diagram illustrating a method of operating a print component, according to one example.

(10) Throughout the drawings, identical reference numbers designate similar, but not necessarily identical, elements. The figures are not necessarily to scale, and the size of some parts may be exaggerated to more clearly illustrate the example shown. Moreover the drawings provide examples and/or implementations consistent with the description; however, the description is not limited to the examples and/or implementations provided in the drawings.

DETAILED DESCRIPTION

(11) In the following detailed description, reference is made to the accompanying drawings which form a part hereof, and in which is shown by way of illustration specific examples in which the disclosure may be practiced. It is to be understood that other examples may be utilized and structural or logical changes may be made without departing from the scope of the present disclosure. The following detailed description, therefore, is not to be taken in a limiting sense, and the scope of the present disclosure is defined by the appended claims. It is to be understood that features of the various examples described herein may be combined, in part or whole, with each other, unless specifically noted otherwise.

(12) Examples of fluidic dies may include fluid actuators. The fluid actuators may include thermal resistor based actuators (e.g. for firing or recirculating fluid), piezoelectric membrane based actuators, electrostatic membrane actuators, mechanical/impact driven membrane actuators, magneto-strictive drive actuators, or other suitable devices that may cause displacement of fluid in response to electrical actuation. Fluidic dies described herein may include a plurality of fluid actuators, which may be referred to as an array of fluid actuators. An actuation event may refer to singular or concurrent actuation of fluid actuators of the fluidic die to cause fluid displacement. An example of an actuation event is a fluid firing event whereby fluid is jetted through a nozzle.

(13) In example fluidic dies, the array of fluid actuators may be arranged in sets of fluid actuators, where each such set of fluid actuators may be referred to as a “primitive” or a “firing primitive.” The number of fluid actuators in a primitive may be referred to as a size of the primitive. In some examples, the set of fluid actuators of each primitive are addressable using a same set of actuation addresses, with each fluid actuator of a primitive corresponding to a different actuation address of the set of actuation addresses, with the addresses being communicated via an address bus. In some examples, a fluidic actuator of a primitive will actuate (e.g., fire) in response to a fire signal (also referred to as a fire pulse) based on actuation data corresponding to the primitive (sometimes also referred to as nozzle data or primitive data) when the actuation address corresponding to the fluidic actuator is present on the address bus.

(14) In some cases, electrical and fluidic operating constraints of a fluidic die may limit which fluid actuators of each primitive may be actuated concurrently for a given actuation event. Primitives facilitate addressing and subsequent actuation of fluid actuator subsets that may be concurrently actuated for a given actuation event to conform to such operating constraints.

(15) To illustrate by way of example, if a fluidic die comprises four primitives, with each primitive

including eight fluid actuators (with each fluid actuator corresponding to a different address of a set of addresses 0 to 7), and where electrical and fluidic constraints limit actuation to one fluid actuator per primitive, a total of four fluid actuators (one from each primitive) may be concurrently actuated for a given actuation event. For example, for a first actuation event, the respective fluid actuator of each primitive corresponding to address “0” may be actuated. For a second actuation event, the respective fluid actuator of each primitive corresponding to address “5” may be actuated. As will be appreciated, such example is provided merely for illustration purposes, with fluidic dies contemplated herein may comprise more or fewer fluid actuators per primitive and more or fewer primitives per die.

(16) Example fluidic dies may include fluid chambers, orifices, and/or other features which may be defined by surfaces fabricated in a substrate of the fluidic die by etching, microfabrication (e.g., photolithography), micromachining processes, or other suitable processes or combinations thereof. Some example substrates may include silicon based substrates, glass based substrates, gallium arsenide based substrates, and/or other such suitable types of substrates for microfabricated devices and structures. As used herein, fluid chambers may include ejection chambers in fluidic communication with nozzle orifices from which fluid may be ejected, and fluidic channels through which fluid may be conveyed. In some examples, fluidic channels may be microfluidic channels where, as used herein, a microfluidic channel may correspond to a channel of sufficiently small size (e.g., of nanometer sized scale, micrometer sized scale, millimeter sized scale, etc.) to facilitate conveyance of small volumes of fluid (e.g., picoliter scale, nanoliter scale, microliter scale, milliliter scale, etc.).

(17) In some examples, a fluid actuator may be arranged as part of a nozzle where, in addition to the fluid actuator, the nozzle includes an ejection chamber in fluidic communication with a nozzle orifice. The fluid actuator is positioned relative to the fluid chamber such that actuation of the fluid actuator causes displacement of fluid within the fluid chamber that may cause ejection of a fluid drop from the fluid chamber via the nozzle orifice. Accordingly, a fluid actuator arranged as part of a nozzle may sometimes be referred to as a fluid ejector or an ejecting actuator.

(18) In some examples, a fluid actuator may be arranged as part of a pump where, in addition to the fluidic actuator, the pump includes a fluidic channel. The fluidic actuator is positioned relative to a fluidic channel such that actuation of the fluid actuator generates fluid displacement in the fluid channel (e.g., a microfluidic channel) to convey fluid within the fluidic die, such as between a fluid supply and a nozzle, for instance. An example of fluid displacement/pumping within the die is sometimes also referred to as micro-recirculation. A fluid actuator arranged to convey fluid within a fluidic channel may sometimes be referred to as a non-ejecting or microrecirculation actuator. In one example nozzle, the fluid actuator may comprise a thermal actuator, where actuation of the fluid actuator (sometimes referred to as “firing”) heats the fluid to form a gaseous drive bubble within the fluid chamber that may cause a fluid drop to be ejected from the nozzle orifice. As described above, fluid actuators may be arranged in arrays (such as columns), where the actuators may be implemented as fluid ejectors and/or pumps, with selective operation of fluid ejectors causing fluid drop ejection and selective operation of pumps causing fluid displacement within the fluidic die. In some examples, the array of fluid actuators may be arranged into primitives.

(19) Some printheads receive data in the form of data packets, sometimes referred to as fire pulse groups or a fire pulse group data packets, where each data packet includes a head portion and a body portion. In some examples, the head portion includes a sequence of start bits and configuration data for on-die functions such as address bits for address drivers, and fire pulse data for fire pulse selection, for example. The body portion of the packet includes primitive data, such as actuator data and/or memory data, that selects which nozzles corresponding to address represented by the address bits in the primitives will be actuated (or fired) and, in some examples, represents data to be written to memory elements of memory arrays associated the primitives. The fire pulse group data pack concludes with stop bits indicating the end of the data packet.

(20) Such printheads include data parsers which use a free-running clock and operate to capture incoming data bits as they are received by the printhead in order to detect the start pattern and thereby identify the beginning of a fire pulse group data packet. Upon detecting a start pattern, the data parser circuitry collects bits as they are received and directs them to the appropriate primitives. In some examples, to determine when the data packet is complete, the data parser circuitry counts the total number of bits received. When the correct number of bits for a data packet has been received, the data parser circuitry stops distributing bits and returns to monitoring incoming data to identify a start sequence for another data packet.

(21) Among other functions, data parser circuitry typically includes several counters, such as to indicate a particular group of primitives to which the data is to be directed (e.g., a printhead may include multiple columns of primitives), and to count a total number bits which have been received, for example. Data parser circuitry consumes relatively large amounts of silicon area on a printhead die, thereby increasing the size and cost of the die. Additionally, data parser circuitry is inflexible and requires each fire pulse group data packet for a printhead to have a fixed length. Additionally, a free running clock can potentially introduce electromagnetic interference (EMI) issues to the die.

(22) The present disclosure, as will be described in greater detail herein, provides a print component having an array of memory elements to serially receive a segment of data bits including configuration data and primitive data each time an intermittent clock signal is received on a clock pad, which eliminates data parser circuitry and a free running clock. Such an arrangement reduces silicon area requirements, eliminates EMI introduced by a free-running clock signal, and enables arrays of fluid actuators having different primitive sizes, such as different fluidic dies, to share a clock and fire signals, which reduces interconnect complexity.

(23) FIG. 1 is a block and schematic diagram generally illustrating a print component **30**, according to one example of the present disclosure, including a plurality of data pads **32**, illustrated as data pads **32-1** to **32-N**, a clock pad **34** to receive an intermittent clock signal **35**, and a plurality of actuator groups **36**, illustrated as actuator groups **36-1** to **36-N**, with each actuator group **36** corresponding to a different one of the data pads **32**. In one example, each of the actuator groups **36** corresponds to a different fluid type. For instance, in one case, print component **30** comprises a printhead with each actuator group corresponding to a different type of ink (e.g., black, cyan, magenta, and yellow). In one example, each actuator group **36** of print component **30** is implemented in a different respective fluidic die where, in one case, each respective fluidic die corresponds to a different liquid type.

(24) According to one example, each actuator group **36** includes a group of configuration functions **38**, illustrated as **38-1** to **38-N**, an array of fluid actuators **40**, illustrated as arrays **40-1** to **40-N**, and an array of memory elements **50**, illustrated as arrays **50-1** to **50-N**. In one case, each group of configuration functions **38** includes a number of configuration functions, illustrated as configuration functions CF(1) to CF(m), for configuring an operational setup of the corresponding actuator group **36**. In examples, configuration functions CF(1) to CF(m) may include functions such as an address driver, a fire pulse configuration function, and a sensor configuration function (e.g., thermal sensors), for instance.

(25) In one example, each array of fluid actuators **40** includes a number of fluid actuators (FAs), with array **40-1** of actuator group **36-1** including fluid actuators FA(1) to FA(x), array **40-2** of actuator group **36-2** including fluid actuators FA(1) to FA(y), and array **40-N** of actuator group **40-N** including fluid actuators FA(1) to FA(z). In one case, each array of fluid actuators **40** may have a same number of fluid actuators ($x=y=z$). In other cases, the arrays of fluid actuators **40** may have differing numbers of fluid actuators ($x \neq y \neq z$).

(26) The array of memory elements **50** of each actuator group **36** comprises a number of memory elements **51**, with each array **50** having a first portion of memory elements **52**, illustrated as first portions **52-1** to **52-N**, corresponding to the respective group of configuration functions **38**, and a second portion of portion of memory elements **54**, illustrated as second portions **56-1** to **56-N**,

corresponding to the respective array of fluid actuators **40**. In some cases, the array of memory elements **50** of each actuator group **36** may have a same number of memory elements **51**. In other cases, the array of memory elements **50** of different actuator groups **36** may have different numbers of memory elements **51**.

(27) The array of memory elements **50** of each actuator group **36** is connected to the corresponding data pad **32** via a corresponding communication path **52**, with the arrays of memory elements **50-1** to **50-N** being respectively connected to data pads **32-1** to **32-N** by communication paths **52-1** to **52-n**. In one example, as illustrated by the arrangement of FIG. 1, each array of memory elements **50** of each group of fluid actuators **36** is connected to and receives intermittent clock signal **35** via clock pad **34**.

(28) In one example, each time intermittent clock **35** is present on clock pad **34** of print component **30**, the array of memory elements **50** of each actuator group **36** serially loads a data segment **33** comprising a series of data bits from the corresponding data pad **32**, illustrated as data segments **33-1** to **33-n**, with the data bits loaded into the first portion of memory elements **52** and into the second portion of memory elements **54** respectively corresponding to the group of configuration functions **38** and to the array of fluid actuators **40**. In one example, each time intermittent clock signal **35** is present on clock pad **34**, the array of memory elements **50** of each actuator group **36** serially loads the series of data bits of a current data segment **33**, which replace the previously loaded data bits of the preceding data segment **33**.

(29) In one example, as will be described in greater detail below (e.g., see FIG. 3), the series of data bits of each data segment **33** include fire pulse groups similar to that described above. However, because print component **30**, loads each data segment **33** only when intermittent clock signal **35** is present on clock pad **34** (i.e., does not employ a free running clock), the fire pulse groups of data segments **33** do not include a start-bit sequence. Since data segments **33** do not include a start-bit sequence and are loaded into the array of memory elements **50** only when intermittent clock signal **35** is present on clock pad **34**, print component **30** and actuator groups **36**, in accordance with the present disclosure, do not include data parser circuitry, thereby saving circuit area and reducing costs.

(30) Additionally, as described in greater detail below, using an intermittent clock signal **35** and an array of memory elements **50** to serially receive data enables print component **30** to support multiple arrays of fluid actuators **40** having differing numbers of fluid actuators and using fire pulse groups of varying lengths while operating on a same intermittent clock signal **35** and sharing a common fire signal (as will be described in greater detail below). Furthermore, employing an intermittent clock signal eliminates potential EMI problems associated with free-running clocks.

(31) FIG. 2 is a block and schematic diagram generally illustrating a print component **30**, according to one example of the present disclosure. In one example, the actuator groups **36-1** to **36-n** are implemented as fluidic dies **37-1** to **37-n**. According to the example of FIG. 2, the fluid actuators (FA) of each of the arrays of fluid actuators **40-1** to **40-n** of actuator groups **36-1** to **36-n** are arranged to form a number of primitives, with the fluid actuators of array **40-1** of actuator group **36-1** arranged to form primitive P(1) to P(x), the fluid actuators of array **40-2** of actuator group **36-2** arranged to form primitive P(1) to P(y), and the fluid actuators of array **40-n** of actuator group **36-n** arranged to form primitive P(1) to P(z), with each primitive including a number of fluid actuators FA(1) to FA(p). In one case, each array of fluid actuators **40** may have a same number of primitives ($x=y=z$). In other cases, the arrays of fluid actuators **40** may have differing numbers of primitives ($x \neq y \neq z$). Although the primitives of each actuator group **36** is illustrated as having a same number of fluid actuators, p, in other examples, the number of fluid actuators in each primitive may vary between actuator groups **36**.

(32) In one example, as illustrated, the array of memory elements **50** of each actuator group **37** comprises a series or chain of memory elements **51** implemented to function as a serial-to-parallel data converter, with first portion **54** of memory elements **51** corresponding to the group of

configuration functions **38**, and second portion of memory elements **56** corresponding to the array of fluid actuators **40**, with each memory element **51** of the second portion **56** corresponding to a different one of the primitives $P(1)$ to $P(x)$. In one example, the array of memory elements **50** of each actuator group **36** comprises a sequential logic circuit (e.g., flip-flop arrays, latch arrays, etc.). In one example, the sequential logic circuit is adapted to function as a serial-in, parallel-out shift register.

(33) According to one example, the group of configuration functions **38** of each actuator group **36** includes an address driver **60**, illustrated as address drivers **60-1** to **60-n**, which drives an address onto a corresponding address bus **62**, illustrated as address buses **62-1** to **62-n**, based on address bits in corresponding memory elements **51** of first portion **54** of the array of memory elements **50**, with memory bus **62** communicating the driven address to fluid actuators $FA(1)$ to $FA(p)$ of each of the corresponding primitives. In one example, print component **30** includes a fire pad **70** to receive a fire signal **72** which is communicated to each of the actuator groups **36** via a communication path **74**.

(34) An example of the operation of print component **30** of FIG. 2 is described below with reference to FIGS. 3 and 4. FIG. 3 is a block and schematic diagram generally illustrating portions of a primitive arrangement for the primitives of actuator groups **36-1** to **36-n** of FIG. 2. For illustrative purposes, the block and schematic diagram of FIG. 2 is described with reference to primitive $P(1)$ of actuator group **36-1** of FIG. 2.

(35) In example, each fluid actuator, illustrated as a thermal resistor in FIG. 3, is connectable between a power source, V_{PP} , and a reference potential (e.g., ground) via a corresponding controllable switch, such as illustrated by FETs **80**.

(36) According to one example, each primitive, including primitive $P(1)$, includes an AND-gate **82** receiving, at a first input, primitive data (e.g., actuator data) for primitive $P(1)$ stored in a local memory element **84**, where local memory element receives such primitive data from corresponding memory element **51** of the array of memory elements **50-1** of actuator group **36-1**. At a second input, AND-gate **82** receives fire signal **72** via communication path **70**. In one example, fire signal **72** is delayed by a delay element **86**, with each primitive having a different delay so that firing of fluid actuators is not simultaneous among primitives $P(1)$ to $P(x)$.

(37) In one example, each fluid actuator has a corresponding address decoder **88** receiving the address driven by address driver **60-1** on address bus **62-1**, and an AND-gate **90** for controlling a gate of FET **80**. AND-gate **90** receives the output of corresponding address decoder **88** at a first input, and the output of AND-gate **82** at a second input. It is noted that address decoder **88** and AND-gate **90** are repeated for each fluid actuator, while AND-gate **82**, memory element **84**, and delay element **86** are repeated for each primitive.

(38) FIG. 4A is a block diagram generally illustrating example data segments **33-1** to **33-n** respectively received by print component **30** via data pads **32-1** to **32-n**. As illustrated, each data segment **33** includes a fire pulse group **100** including a first portion of data bits **102** corresponding to the group of configuration functions **38** (sometimes referred to as configuration data), and a second portion of data bits **104** corresponding to the array of fluid actuators **40** (sometimes referred to as primitive data). For instance, with respect to data segment **33-1**, the data bits of the first portion of data bits **102-1** correspond to the group of configuration functions **38-1** and include address data bits for address driver **60-1**, and the data bits of the second portion of data bits **104-1** correspond to the array of fluid actuators **40-1**, with each data bit of second portion **104-1** corresponding to a different one of the primitives $P(1)$ to $P(x)$. For each data segment **33**, the number of data bits of the fire pulse group **32** (i.e., the number of fire pulse bits) is equal to the sum of the number of bits of the first portion of data bits **102** (i.e., configuration data bits) and the number of bits of the second portion of data bits **104** (i.e., primitive data).

(39) According to the example of FIG. 4A, second portion **104-1** of fire pulse group **100-1** of data segment **33-1** is illustrated as having more primitive data bits than second portion **104-2** of fire

pulse group **100-2** of data segment **33-2**, and second portion **104-2** of fire pulse group **100-2** of data segment **33-2** is illustrated as having more primitive data bits than second portion **104-n** of fire pulse group **100-n** of data segment **33-n**, meaning that, with reference to FIG. 2, the array of fluidic actuators **40-1** of fluidic die **36-1** has a greater number of primitives than the array of fluidic actuators **40-2** of fluidic die **36-2**, while the array of fluidic actuators **40-2** of fluidic die **36-2** has a greater number of primitives than the array of fluidic actuators **40-n** of fluidic die **36-n** (i.e., $x > y > z$). As a result, fire pulse group **100-1** has more fire pulse group bits than fire pulse group **100-2**, and fire pulse group **100-2** has more fire pulse group bits than fire pulse group **100-n**, meaning that data segment **33-1** is longer (i.e., has more data segment bits) than data segment **33-2**, and that data segment **33-2** is longer (i.e., has more data segment bits) than data segment **33-n**.

(40) With reference to FIG. 2, upon intermittent clock signal **35** being received at clock pad **34** (e.g., upon receiving the first rising edge of intermittent clock signal **35**), data segments **33-1** to **33-n** are serially loaded into the memory elements **51** of their respective arrays of memory elements **50-1** to **50-n** of actuator groups **36-1** to **36-n**. However, when sharing a same intermittent clock signal **35**, as illustrated by the example implementation of FIG. 2, because of their different lengths, the number of cycles of intermittent clock signal **35** needed to load fire pulse group **100-1** of data segment **33-1** into array of memory elements **50-1** is greater than a number of clock cycles needed to load fire pulse groups **100-2** and **100-n** of data segments **33-2** and **33-n** into their respective arrays of memory elements **50-2** and **50-n**. As a result, data bits of fire pulse groups **100-2** and **100-n** of data segments **33-2** and **33-n** will begin being respectively shifted out of arrays of memory elements **50-2** and **50-n** before data bits of fire pulse group **100-1** of data segment **33-1** have finished being serially loaded into the array of memory elements **50-1**. Consequently, if not accounted for, incorrect data will be populating the memory elements of arrays **50-2** and **50-n** upon completion of loading data segment **33-1** into array **50-1**.

(41) With reference to FIG. 4B, according to one example, when sharing an intermittent clock signal, such as clock signal **35**, in order to make each of the data segments **33-1** to **33-n** equal in length (i.e., a same number of bits) so as to take a same number of clock cycles of intermittent clock signal **35** to load into their respective memory arrays **50-1** to **50-n**, in addition to fire pulse groups **100-2** and **100-n**, data segments **33-1** and **33-n** each include a pre-pended segment of filler bits **110-1** and **110-n**. According to one example, as illustrated, since data segment **33-1** is the longest data segment (i.e., has the most segment bits), the segment of filler bits **110-1** of data segment **33-1** contains no filler bits, while segments of filler bits **110-2** and **110-n** each have a number of filler bits to respectively make data segments **33-2** and **33-n** the same length as data segment **33-1** (with filler bit segment **33-n** having more filler bits than filler bit segment **33-2**). According to the example illustration of FIG. 4B, in general, segments of filler bits **110** are added to each shorter data segment **33** of data segments **33-1** to **33-n** so that all data segments **33-1** to **33-n** have a length the same as the longest data segment **33** of data segments **33-1** to **33-n**.

(42) By pre-pending filler bit segments **110-1** to **110-n** to data segments **33-1** and **33-n**, in a case where an intermittent clock signal is shared by actuator groups **36-1** to **36-n**, when serially loading data segments **33-1** to **33-n** into their respective arrays of memory elements **50-1** to **50-n**, the last data bit of each data segments **33-1** to **33-n** will be loaded on the same clock cycle so that each fire pulse group is properly loaded into their respective memory array **50-1** to **50-n**, with the first and second portions of data bits **102** and **104** being respectively loaded into first and second portions **54** and **56** of the corresponding array of memory elements **50**.

(43) Prepending filler bit segments **110** to at least data segments **33** having shorter lengths so that all data segments **33** have a same length enables a clock signal **35** to be shared by multiple arrays of fluidic actuators **36** even when such arrays of fluidic actuators **36** have differing numbers of fluid actuators (FAs), which reduces and simplifies circuitry, such as that of print component **30**.

(44) In some examples, each of the data segments **33-1** to **33-n** includes a filler bit segment **100** including a number of filler bits, where the number of filler bits in each filler bit segment **100-1** to

100-n is such that each of the data segments **33-1** to **33-n** has a same length. In one example, each of the filler bits has either a logic “high” value (e.g. “1”) or a logic “low” value (“0”), where the filler bits of each filler bit segment **100** have a pattern of logic “low” and logic “high” values to mitigate electromagnetic effects on print component **30** as data segments **33-1** to **33-n** are respectively serially loaded in memory arrays **50-1** to **50-n**.

(45) Continuing with the illustrative example above, referring to FIGS. 2-3, in one case, upon the final data bit of each of the data segments **33-1** to **33-n** being loaded into the respective array of memory elements **50-1** to **50-n** (e.g., the last data bit each of the second portions **104-1** to **104-n** of fire pulse groups **100-1** to **100-n** being loaded into their respective memory element **51** corresponding to primitive P(1)), intermittent clock signal **35** is removed from clock pad **34** so that serial loading of data into memory arrays **50-1** to **50-n** ceases.

(46) According to one example, upon completion of loading of fire pulse groups **100-1** to **100-n** into their respective memory arrays **50-1** to **50-n**, a fire signal **72** (e.g., a fire pulse signal) is received on fire pad **70**. With reference to FIGS. 2 and 3, in one example, in response to receipt of fire pulse signal **72**, data stored in each memory element **51** of each array of memory elements **50-1** to **50-n** are parallel shifted into a corresponding memory element in the corresponding array of fluid actuators **40-1** to **40-n** or the group configuration functions **38-1** to **38-n**. For example, in FIG. 3, in response to fire signal **72**, primitive data stored in memory element **51** is shifted to a corresponding memory element **84** in primitive P(1).

(47) In one example, after being parallel shifted out of the arrays of memory elements **50-1** to **50-n**, the fire pulse group data is processed by the corresponding groups of configuration functions **38-1** to **38-n** and primitives (P(1) to P(x), P(1) to P(y), and P(1) to P(z)) to operate selected fluid actuators (FAs) to circulate fluid or eject fluid drops. For instance, with reference to FIG. 3, in one example, if the primitive data stored in memory element **84** has a logic high (e.g., “1”) and a fire pulse signal **72** is present on communication path **74**, the output of AND-gate **82** is set to a logic “high”. If the address driven on address bus **62-1** by address encoder **60-1** in response to the address bits received from the corresponding memory element of the second group of memory elements **54-1** represents address “0”, the output of Address Decoder “0” **88** is set to a logic “high”. With the output of AND-gate **82** and Address Decoder “0” **88** each set to a logic “high”, the output of AND-gate **90** is also set to a logic “high”, thereby turning “on” corresponding FET **80** to energize fluid actuator FA(0) to displace fluid (e.g., eject a fluid drop).

(48) In one example, upon the fire pulse group data being shifted out of the arrays of memory elements **50-1** to **50-n** in response to fire signal **72**, intermittent clock signal **35** is again received via clock pad **34** and next data segments **33-1** to **33-n** are serially loaded into the arrays of memory elements **50-1** to **50-n**.

(49) FIG. 5 is a block and schematic diagram generally illustrating print component **30** of FIG. 2, where in addition to fluid actuators FA(1) to FA(p), primitives P(1) to P(x), P(1) to P(y), and P(1) to P(z) of actuator groups **40-1** to **40-n** each include an array of memory elements, respectively illustrated as M(1) to M(x), M(1) to M(y), and M(1) to M(z). In one example, as illustrated, each of the groups of configurations **38-1** to **38-n** may include one or more memories, CM, each corresponding to a different one of the configuration functions.

(50) In one example, print component **30** of FIG. 5 further includes a mode pad **78** to receive a mode signal **79**. In one example, based on a state of mode signal **79**, upon fire signal **72** being raised on fire pad **70**, rather than data stored in the array of memory elements **50-1** to **50-n** being shifted to the fluid actuators and configuration functions, the data is shifted to the primitive memory arrays of their respective primitives (e.g. M(1) to M(x), M(1) to M(y), and M(1) to M(z)) and to the configuration memory, CM, of the respective group of configuration functions **38-1** to **38-n**.

(51) FIG. 6 is a block and schematic diagram generally illustrating print component **30** of FIG. 5, where, in lieu of fluidic dies **37-1** to **37-n** sharing a common intermittent clock signal **35**, each

fluidic die **37-1** to **37-n** receives its own corresponding intermittent clock signal, illustrated as clock signals **35-1** to **35-n** via corresponding clock pads **34-1** to **34-n**. With reference to FIGS. **2-4**, since intermittent clock signals **35-1** to **35-n** may be separately controlled (e.g., may start and/or stop at differing times), data segments **33-1** to **33-n** do not need to be of a same length and, thus, may not include filler bit segments **110**. Referring to FIG. **6**, upon completion of loading of fire pulse groups **100-1** to **100-n** of data segments **33-1** to **33-n** into the array of memory elements **50-1** to **50-n** of the corresponding fluid die **37-1** to **37-n**, fire signal **72** may be raised to initiate operations on the fire pulse group data (as described above).

(52) FIG. **7** is a block diagram illustrating one example of a fluid ejection system **200**. Fluid ejection system **200** includes a fluid ejection assembly, such as printhead assembly **204**, and a fluid supply assembly, such as ink supply assembly **216**. In the illustrated example, fluid ejection system **200** also includes a service station assembly **208**, a carriage assembly **222**, a print media transport assembly **226**, and an electronic controller **230**. While the following description provides examples of systems and assemblies for fluid handling with regard to ink, the disclosed systems and assemblies are also applicable to the handling of fluids other than ink.

(53) Printhead assembly **204** includes at least one printhead **212** which ejects drops of ink or fluid through a plurality of orifices or nozzles **214**, where printhead **212** may be implemented, in one example, as print component **30** with fluid actuators (FAs) of actuator groups **36-1** to **36-n** implemented as nozzles **214**, as previously described herein by FIG. **2**, for instance. In one example, the drops are directed toward a medium, such as print media **232**, so as to print onto print media **232**. In one example, print media **232** includes any type of suitable sheet material, such as paper, card stock, transparencies, Mylar, fabric, and the like. In another example, print media **232** includes media for three-dimensional (3D) printing, such as a powder bed, or media for bioprinting and/or drug discovery testing, such as a reservoir or container. In one example, nozzles **214** are arranged in at least one column or array such that properly sequenced ejection of ink from nozzles **214** causes characters, symbols, and/or other graphics or images to be printed upon print media **232** as printhead assembly **204** and print media **232** are moved relative to each other.

(54) Ink supply assembly **216** supplies ink to printhead assembly **204** and includes a reservoir **218** for storing ink. As such, in one example, ink flows from reservoir **218** to printhead assembly **204**. In one example, printhead assembly **204** and ink supply assembly **216** are housed together in an inkjet or fluid-jet print cartridge or pen. In another example, ink supply assembly **216** is separate from printhead assembly **204** and supplies ink to printhead assembly **204** through an interface connection **220**, such as a supply tube and/or valve.

(55) Carriage assembly **222** positions printhead assembly **204** relative to print media transport assembly **226**, and print media transport assembly **226** positions print media **232** relative to printhead assembly **204**. Thus, a print zone **234** is defined adjacent to nozzles **214** in an area between printhead assembly **204** and print media **232**. In one example, printhead assembly **204** is a scanning type printhead assembly such that carriage assembly **222** moves printhead assembly **204** relative to print media transport assembly **226**. In another example, printhead assembly **204** is a non-scanning type printhead assembly such that carriage assembly **222** fixes printhead assembly **204** at a prescribed position relative to print media transport assembly **226**.

(56) Service station assembly **208** provides for spitting, wiping, capping, and/or priming of printhead assembly **204** to maintain the functionality of printhead assembly **204** and, more specifically, nozzles **214**. For example, service station assembly **208** may include a rubber blade or wiper which is periodically passed over printhead assembly **204** to wipe and clean nozzles **214** of excess ink. In addition, service station assembly **208** may include a cap that covers printhead assembly **204** to protect nozzles **214** from drying out during periods of non-use. In addition, service station assembly **208** may include a spittoon into which printhead assembly **204** ejects ink during spits to ensure that reservoir **218** maintains an appropriate level of pressure and fluidity, and to ensure that nozzles **214** do not clog or weep. Functions of service station assembly **208** may

include relative motion between service station assembly **208** and printhead assembly **204**.

(57) Electronic controller **230** communicates with printhead assembly **204** through a communication path **206**, service station assembly **208** through a communication path **210**, carriage assembly **222** through a communication path **224**, and print media transport assembly **226** through a communication path **228**. In one example, when printhead assembly **204** is mounted in carriage assembly **222**, electronic controller **230** and printhead assembly **204** may communicate via carriage assembly **222** through a communication path **202**. Electronic controller **230** may also communicate with ink supply assembly **216** such that, in one implementation, a new (or used) ink supply may be detected.

(58) Electronic controller **230** receives data **236** from a host system, such as a computer, and may include memory for temporarily storing data **236**. Data **236** may be sent to fluid ejection system **200** along an electronic, infrared, optical or other information transfer path. Data **236** represent, for example, a document and/or file to be printed. As such, data **236** form a print job for fluid ejection system **200** and includes at least one print job command and/or command parameter.

(59) In one example, electronic controller **230** provides control of printhead assembly **204** including timing control for ejection of ink drops from nozzles **214**. As such, electronic controller **230** defines a pattern of ejected ink drops which form characters, symbols, and/or other graphics or images on print media **232**. Timing control and, therefore, the pattern of ejected ink drops, is determined by the print job commands and/or command parameters. In one example, logic and drive circuitry forming a portion of electronic controller **230** is located on printhead assembly **204**. In another example, logic and drive circuitry forming a portion of electronic controller **230** is located off printhead assembly **204**. In another example, logic and drive circuitry forming a portion of electronic controller **230** is located off printhead assembly **204**. In one example, data segments **33-1** to **33-n**, intermittent clock signal **35**, fire signal **72**, and mode signal **79** may be provided to print component **30** by electronic controller **230**, where electronic controller **230** may be remote from print component **30**.

(60) FIG. **8** is a flow diagram illustrating a method **300** of operating a print component, such as print component **30** of FIGS. **2-4**, in accordance with one example of the present disclosure. At **302**, method **300** includes receiving data segments on a number of data pads, such as receiving data segments **33-1** to **33-n** on data pads **32-1** to **32-n** as illustrated by FIG. **2**, where each data segment comprises a number of segment bits, the number of segment bits including a fire pulse group comprising a number of fire pulse group bits, with the number of segment bits being at least equal to the number of fire pulse group bits, such as illustrated by FIG. **4A** where each data segment **33-1** to **33-n** respectively includes a fire pulse group **100-1** to **100-n**.

(61) At **304**, method **300** includes receiving an intermittent clock signal on a clock pad, such as print component **30** of FIG. **2** receiving an intermittent clock signal **35** on clock pad **34**. At **306**, method **300** includes arranging a number of fluid actuators to form a number of fluid actuator arrays, each array of fluid actuators having a corresponding array of memory elements corresponding to a different one of the data pads, such as actuator groups **36-1** to **36-n** of FIG. **2** respectively including an array of fluid actuators **40-1** to **40-n**, with the arrays of fluid actuators **40-1** to **40-n** respectively having a corresponding array of memory elements **50-1** to **50-n**, with the array of memory elements **50-1** to **50-n** respectively having corresponding data pads **32-1** to **32-n**.

(62) At **308**, method **100** includes serially loading a data segment from the corresponding data pad into each array of memory elements each time the intermittent clock signal is present on the clock pad to store at least the fire pulse group bits, such as respectively loading data segments **33-1** to **33-n** (as illustrated by FIGS. **4A** and **4B**) into arrays of memory elements **50-1** to **50-n** so as to respectively store at least fire pulse segments **100-1** to **100-n**.

(63) Although specific examples have been illustrated and described herein, a variety of alternate and/or equivalent implementations may be substituted for the specific examples shown and described without departing from the scope of the present disclosure. This application is intended

to cover any adaptations or variations of the specific examples discussed herein. Therefore, it is intended that this disclosure be limited only by the claims and the equivalents thereof.

Claims

1. A set of print components, comprising: a plurality of dies, each corresponding to a different liquid type, wherein the plurality of dies share an intermittent clock signal and each die includes: a data pad to receive data segments, each data segment comprising a number of segment bits, the number of segment bits including a fire pulse group comprising a number of fire pulse group bits different for each of the plurality of dies, wherein the number of segment bits is at least equal to the number of fire pulse group bits, and the data segments received on the data pad include a group of filler bits so that the number of segment bits of data segments received by one data pad is the same as a number of segment bits of data segments received by another data pad; a fluidic actuator array corresponding to the intermittent clock signal and to the data pad, having a corresponding array of memory elements; and circuitry, connected to the data pad and the fluidic actuator array, and including the corresponding array of memory elements, the circuitry configured such that a rising edge of the intermittent clock signal causes the corresponding array of memory elements to serially receive a data segment from the data pad and to store at least the fire pulse group bits.
2. The set of print components of claim 1, each fluidic actuator array having a corresponding group of configuration functions.
3. The set of print components of claim 2, each array of memory elements including a first portion of memory elements corresponding to the group of configuration functions and a second portion of memory elements corresponding to the fluidic actuator array.
4. The set of print components of claim 1, the array of memory elements comprising a chain of memory elements adapted to function as a serial-to-parallel data converter.
5. The set of print components of claim 4, wherein the array of memory elements includes a sequential logic circuit.
6. The set of print components of claim 5, the sequential logic circuit adapted to function as a serial-in, parallel-out shift register.
7. The set of print components of claim 1, wherein the corresponding array of memory elements is to load the data segments from the corresponding one of the plurality of data pads only when the intermittent clock signal is present on the clock pad.
8. The set of print components of claim 1, wherein the fire pulse group does not include a start bit sequence that identifies a beginning of the fire pulse group.
9. The set of print components of claim 8, wherein the print component does not include data parser circuitry to detect the beginning of the fire pulse group.
10. A set of print components, comprising: a plurality of dies, each corresponding to a different liquid type, wherein the plurality of dies share an intermittent clock signal and a fire signal, and each die includes: a fire pad to receive the fire signal; a clock pad to provide the intermittent clock signal provided at differing times; at least one data pad to receive data segments, each data segment comprising a number of segment bits, the number of segment bits including a fire pulse group comprising a number of fire pulse bits different for each of the plurality of dies, wherein the number of segment bits is at least equal to the number of fire pulse group bits, and the data segments received on the at least one data pad include a group of filler bits so that the number of segment bits of data segments received by one data pad is the same as a number of segment bits of data segments received by another data pad; a fluidic actuator array corresponding to the intermittent clock signal and to the at least one data pad, having a corresponding array of memory elements; and circuitry, connected to the at least data pad and the fluidic actuator array, and including the corresponding array of memory elements, the circuitry configured such that a rising edge of the intermittent clock signal causes the corresponding array of memory elements to serially

receive a data segment from the at least one data pad and to store the fire pulse bits.

11. The set of print components of claim 10, the array of memory elements comprising a chain of memory elements adapted to function as a serial-to-parallel data converter.

12. The set of print components of claim 11, the array of memory elements comprising a sequential logic circuit.

13. The set of print components of claim 12, the sequential logic circuit adapted to function as a serial-in, parallel-out shift register.

14. The set of print components of claim 10, including: a plurality of clock pads, wherein each clock pad is to receive a corresponding one of a plurality of intermittent clock signals that include the intermittent clock signal, and corresponds to each of the plurality of dies.

15. The set of print components of claim 14, wherein the plurality of intermittent clock signals are provided at differing times.

16. The set of print components of claim 10, wherein the corresponding array of memory elements is to load the data segments from the corresponding one of the plurality of data pads only when the intermittent clock signal is present on the clock pad.

17. The set of print components of claim 16, wherein the corresponding array of memory elements is to serially load the data segments in response to the clock pad receiving a rising edge of the intermittent clock signal.

18. The set of print components of claim 10, wherein the fire pulse group does not include a start bit sequence that identifies a beginning of the fire pulse group.

19. The set of print components of claim 18, wherein the print component does not include data parser circuitry to detect the beginning of the fire pulse group.

20. A plurality of print components, comprising: a fluidic actuator array of each component, each array corresponding to a different liquid type, wherein the plurality of fluidic actuator arrays share an intermittent clock signal and a fire signal, and each component includes: a clock pad to provide the intermittent clock signal; a fire pad to receive the fire signal; at least one data pad to receive data segments, each data segment comprising a number of segment bits, the number of segment bits including a fire pulse group comprising a number of fire pulse group bits, wherein the number of segment bits is at least equal to the number of fire pulse group bits, and the data segments received on the at least one data pad include a group of filler bits so that the number of segment bits of data segments received by one data pad is the same as a number of segment bits of data segments received by another data pad; and circuitry, connected to the at least one data pad and the fluidic actuator array, and including at least one corresponding memory element, wherein a rising edge of the intermittent clock signal causes the at least one corresponding memory element to serially receive a data segment from the at least one data pad and to store the at least one fire pulse group bit.

21. The plurality of print components of claim 20, the at least one corresponding memory element comprising a chain of memory elements adapted to function as a serial-to-parallel data converter.

22. The plurality of print components of claim 21, the at least one corresponding memory element comprising a sequential logic circuit.

23. The plurality of print components of claim 22, the sequential logic circuit adapted to function as a serial-in, parallel-out shift register.

24. The plurality of print components of claim 20, comprising a plurality of data pads, each of which corresponds to one of the plurality of fluidic actuator arrays.

25. The plurality of print components of claim 20, the at least one fluidic actuator array arranged to form a plurality of primitives, each primitive having a same number of fluid actuators, and the at least one corresponding memory element including a first portion corresponding to at least one configuration function and a second portion corresponding to the at least one fluidic actuator array.

26. The plurality of print components of claim 20, the print component comprising a printhead.

27. The plurality of print components of claim 20, the print component comprising a mode pad to

receive a mode signal, wherein a state of the mode signal indicates whether a data value stored in the at least one corresponding memory element corresponds to the at least one fluidic actuator array.
